

matic gearing. Electronics plays a more significant role in automatic-transmission cars. Herein, the automatic transmission of a car is controlled by the transmission control unit (TCU). The TCU collects information from the sensors attached to the vehicle. It further uses the data to do gear shifting at the right time, which helps to increase the car's performance and efficiency.

Multiple sensors are involved in this setup. The common ones include:

1. Vehicle speed sensor. It sends different frequencies to the TCU to determine the rotational speed of the engine. This input from the sensor lets the TCU decide when the gear change should take place.

2. Throttle position sensor. This sensor is connected to the gear shaft, and informs the TCU about the current gear that's been engaged by the driver (or automatically).

3. Turbine speed sensor. It provides real-time information on how the air intake system is functioning at a given time.

4. Transmission fluid temperature sensor. It measures the temperature of oil, fuel, water or engine coolant and provides a signal to the powertrain control module.

The brakes. Anti-lock brake systems (ABS) are becoming increasingly popular in cars. These have two main advantages: they let you stop faster and they let you steer even while you have applied the brakes. The ABS has four major parts, namely, speed sensors, controllers, valves and pump. The first two are assisted by electronics.

The ABS keeps itself updated by taking inputs from the speed sensor located at the wheel, such that it knows when the wheels are about to lock up. The valves are placed in the brake line of each wheel. Whenever they open, the pressure from the cylinder is passed through the brake. Since it is capable of releasing pressure from the valve, a pump is also included in the module to fill the pressure back. The controller keeps itself updated with information of the

sensors and the valve.

The dashboard. Dashboard is the portion with which the driver of a car interacts the most. So what are the features available and what can be added to the dashboard to make it interactive?

The dashboard basically contains panels that show the readings of the different sensors that are used throughout the body of the car. The good-old dashboard has a steering wheel through which the analogue meter could be viewed. It gives indication of the fuel level, the speed at which the car is running, the tachometer and the information regarding the oil level, neutral state of car, etc. With electronics invading into the mechanical arena, we find most of the analogue meter turning digital.

The dashboard of a car can also have the GPS, audio systems, air-conditioner controls, etc. The GPS system can be used by the driver to keep track of his way on the highway. On the dashboard, touch screen can be embedded with an Internet connection. Using it the drivers can keep themselves updated on-the-go. But, with the increasing number of interactive devices, it could be dangerous for the driver to lose sight of the road.

Parking sensors

These are among the latest features being included in Indian cars, especially the high-end ones. You can also buy them off-the-shelf from markets like Palika Bazaar (New Delhi) or Lamington Road (Mumbai) and get them installed.

Basically, a combination of a sensor and a display unit, these indicators warn the driver through beeps and visual representation about how close is an obstruction from the bumper. And, they are not limited to bumpers on the hind side only. Many cars come with sensors on the front bumper too, to assist you in parking your car.

Auto-sensing wipers

If your car has the auto-sensing wiper system, its wipers will turn on the mo-

ment they detect rain. In fact, the speed of your wiper blades increases with the intensity of rain!

A typical setup consists of sensors and infrared light. An infrared beam is reflected off the outside surface of the windshield to the infrared sensor array. When moisture strikes the windshield, the system experiences an interruption to its infrared beam. Advanced analogue and digital signal processing determines the intensity of rain. The sensor, on the basis of the moisture on the screen, determines the speed of the wiper movement.

Auto-sensing lights

If wipers can start off automatically when it rains, why not your headlights when it gets dark? Now they do. Many high-end cars in India come with auto-sensing lighting features too. Here, the headlamps of the car glow automatically when it starts getting dark. The fundamental behind it is the light-dependent resistor (LDR) or optical sensors used in the circuit of the headlamp. So as the light intensity decreases to reach a certain value, the circuit triggers the headlights to turn on.

Safety and security systems

Today, cars are packed with many security and safety features. There are several layers of security that such systems would provide, such as:

1. Security against theft. By means of anti-theft devices, user-recognition electronics and immobilisers

2. Security against car misbehaviour. By means of tyre pressure monitors, temperature sensors, and anti-skid and anti-lock devices

3. Personnel safety in case of collisions. By means of crash sensors, air-bag actuators and warning systems

4. Security against accidents and mishaps. By means of collision-avoidance systems (under evolution)

Desmond Tan, senior marketing manager at Fujitsu Microelectronics, explains, "Embedded electronics enhances the security of the car in areas